Exploring Network Interfaces And Packet Transmission Using OWASP Broken Web Applications

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Executive Summary

This lab focused on exploring and analyzing network interface configuration on a system while monitoring transmission behavior and patterns from different points of observation. Key activities include capturing and analyzing network packets using various tools and understanding the process of packet transmission to and from the OWASP Broken Web Application VM. The lab revealed that connections within and outside a host begins from interfaces activation. Transport layers showed how the host establishes connections through the 3 ways handshakes which is an incredible measure for legitimate communication validation.

1.0 Introduction

The Network interfaces are the physical or virtual connection points that allow a device to send and receive data on a network, such as Ethernet ports, Wi-Fi adapters, or virtual NICs. Each interface has an IP address and a MAC address that identifies it on the network.

Packet transmission on these interfaces is the process of breaking data into small units called packets, adding addressing and control information, and sending them through the network interface across links and routers to the destination, where they are reassembled into the original data.

2.0 Aim and objectives

The aim of this lab is to dive deeper into the concept of network interfaces and packet transmission by analyzing the network interface configuration on the kali Linux system.

You will explore how to capture and analyze network packets using various tools and understand the process of packet

The Objectives of this lab is to

- to capture and analyze network packets using various tools and understand the process of packet transmission to and from the OWASP Broken Web Application VM
- Understand the role and configuration of network interfaces.
- Use packet capture tools to analyze network traffic and understand how data is transmitted.
- gain experience in monitoring and troubleshooting network interfaces and traffic.

2.1 Tools

- Kali Linux
- OWASP BWA
- Wireshark
- Tshark

3.0 Methodology

3.1 Pinging the OWASP BWA

- The Linux terminal was opened and the following command was run
- **ifconfig** and **ip addr show**
- The network interface activated on the machine includes **eth0: 10.0.2.15** and **lo: 127.0.0.1** (loopback address)
- The interface **eth0** is the interface holding the assigned ip address of the machine for connecting to another machine within or outside the network
- The **lo** is a **loopback** address of the device used for testing the correct configuration and arrangement of the **TCP/IP stack** of the network

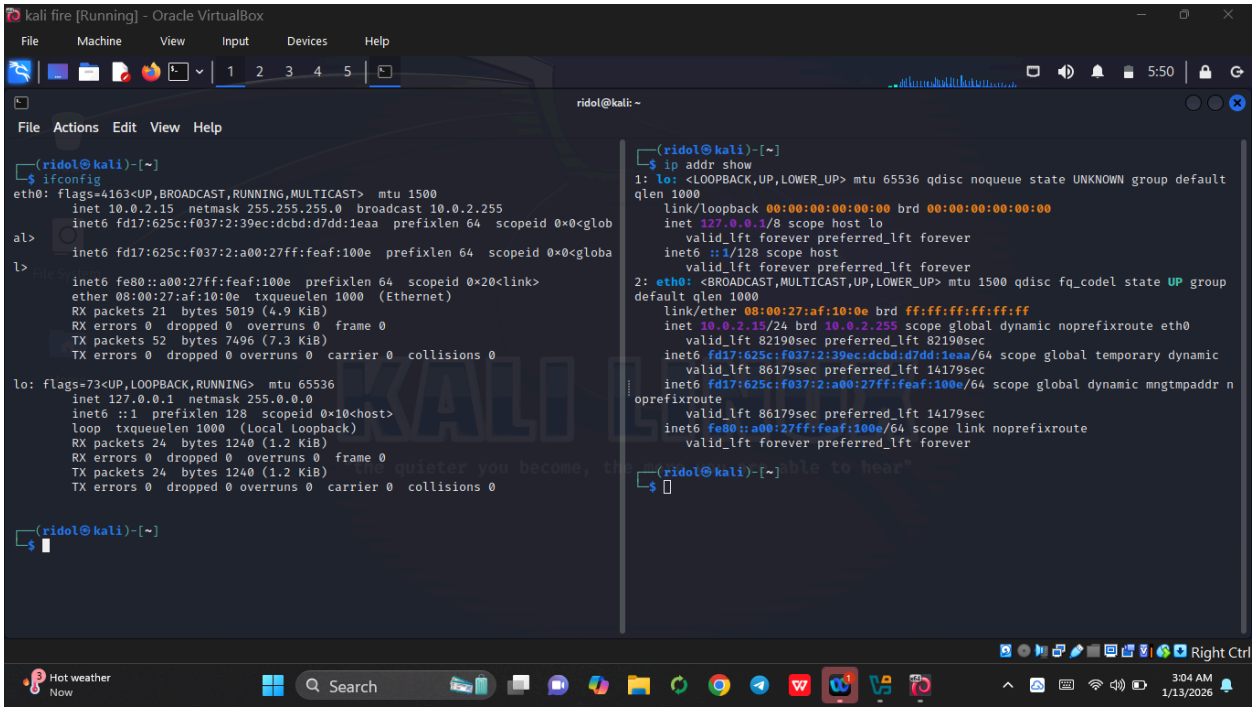

Fig 1: Evidence showing the interferences on the Linux system terminal

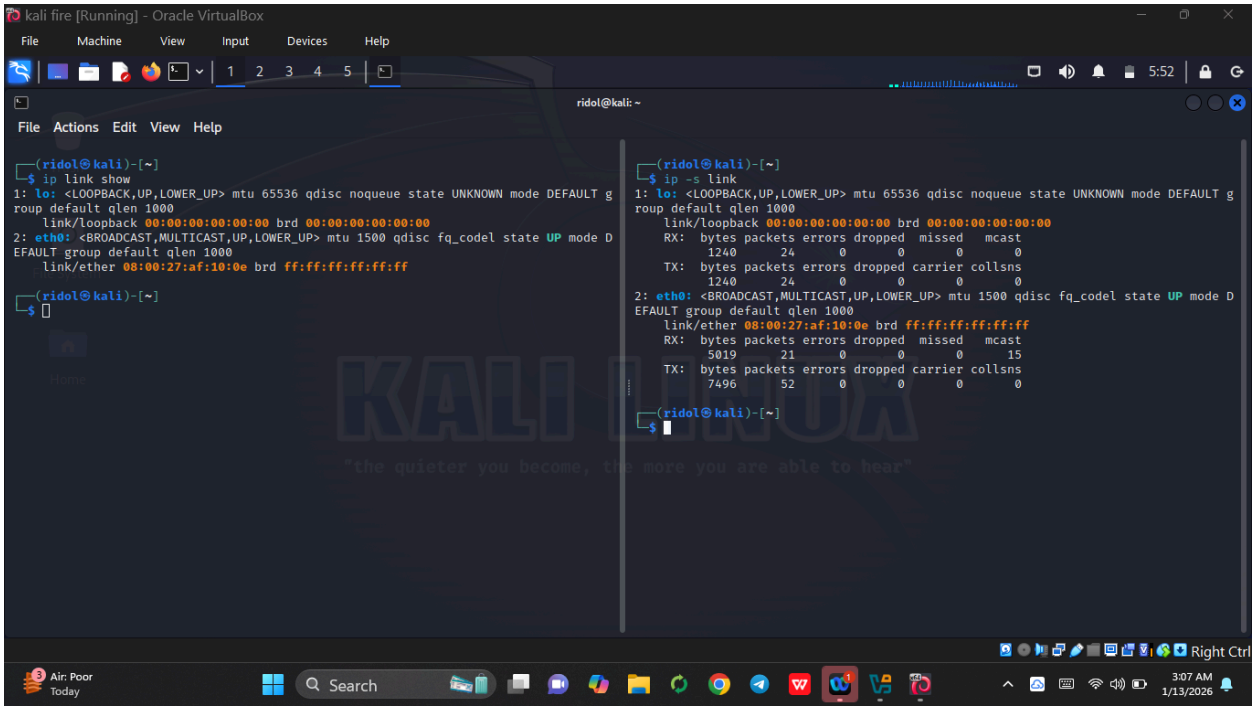

Fig 2: Evidence showing the interferences on the Linux system terminal

3.2 Capturing Packets on Specific Interface

- The Linux terminal was opened and the following command was run
- **Sudo tcpdump -i eth0** and **Sudo tcpdump -i eth0 -w capture.pcap**
- The first command **captured active tcp traffic going on the interface eth0** while the second command **captured and saved in a capture.pcap file** for future reference

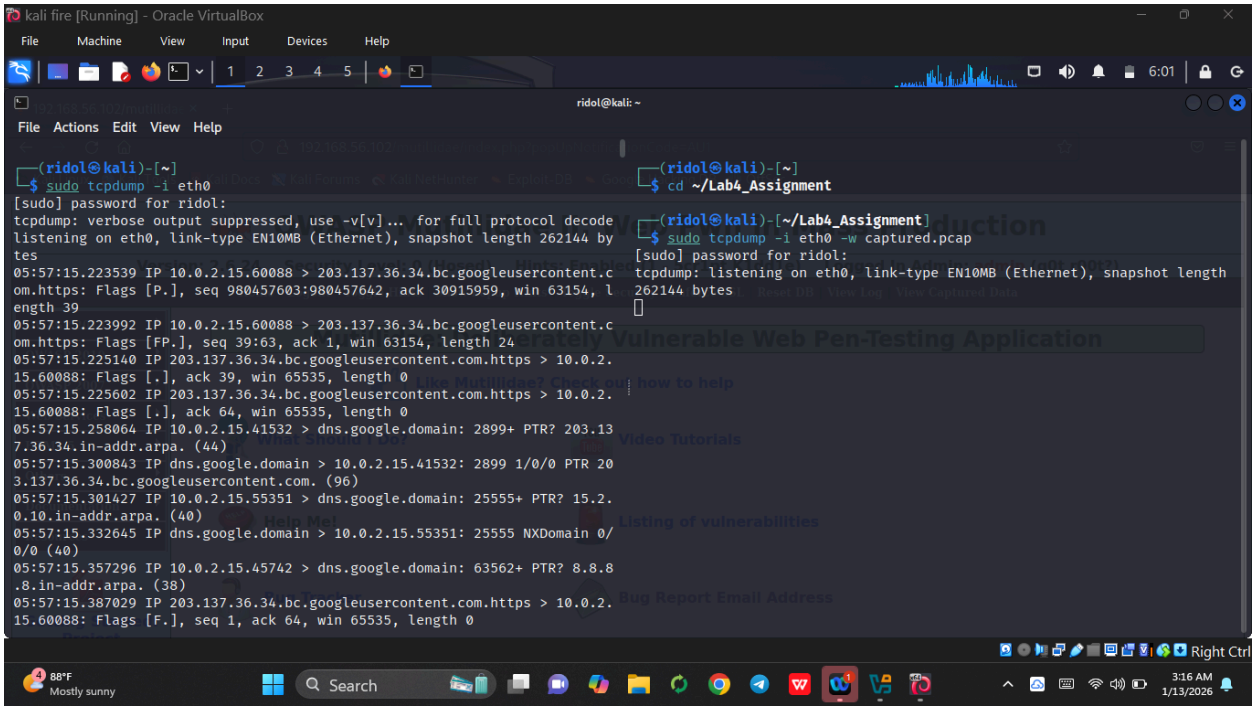

Fig 3: Evidence showing captured traffic on interface eth0

3.3 Examining Network Statistics

- The Linux terminal was opened and the following command was run
- **netstat -i** and **ss -s**
- This showed an established **tcp connection** between the kali Linux and the owasp bwa machine **ip addresses** and the connection **port**
- This indicates a successful 3-ways handshakes on the transport layers **(SYN SYN/ACK ACK)**

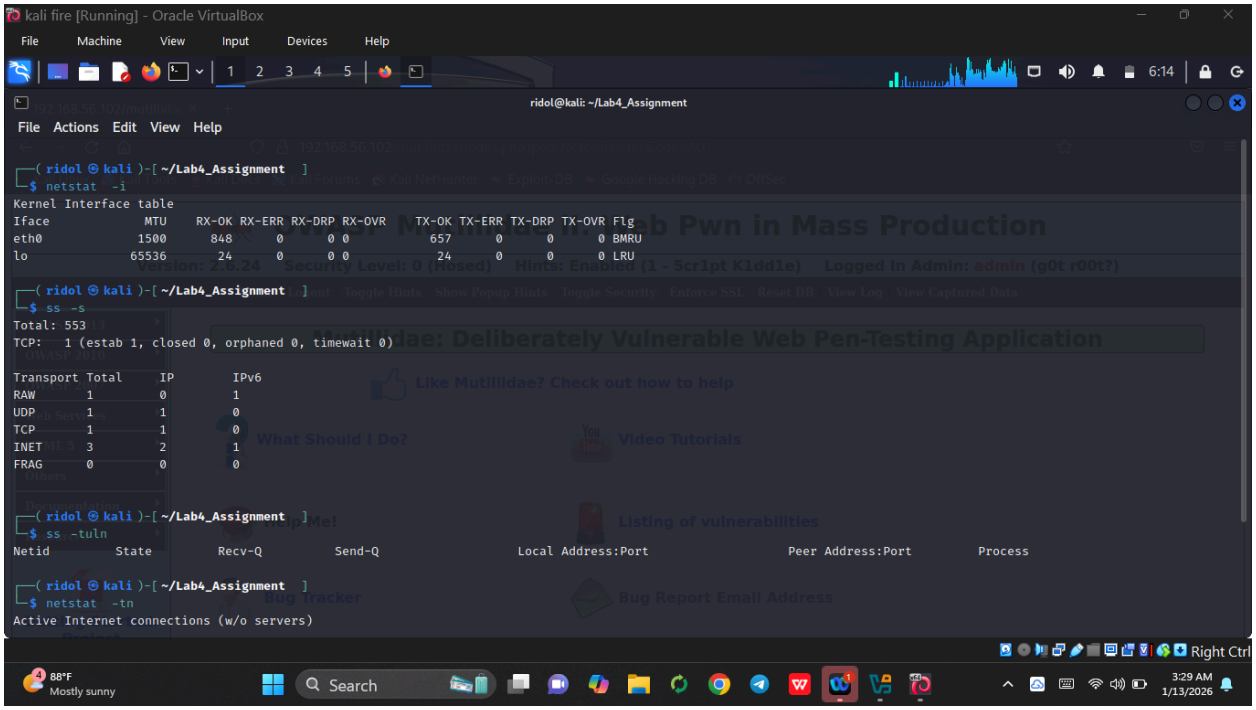

Fig 4:Evidence showing network statistics on the Linux terminal

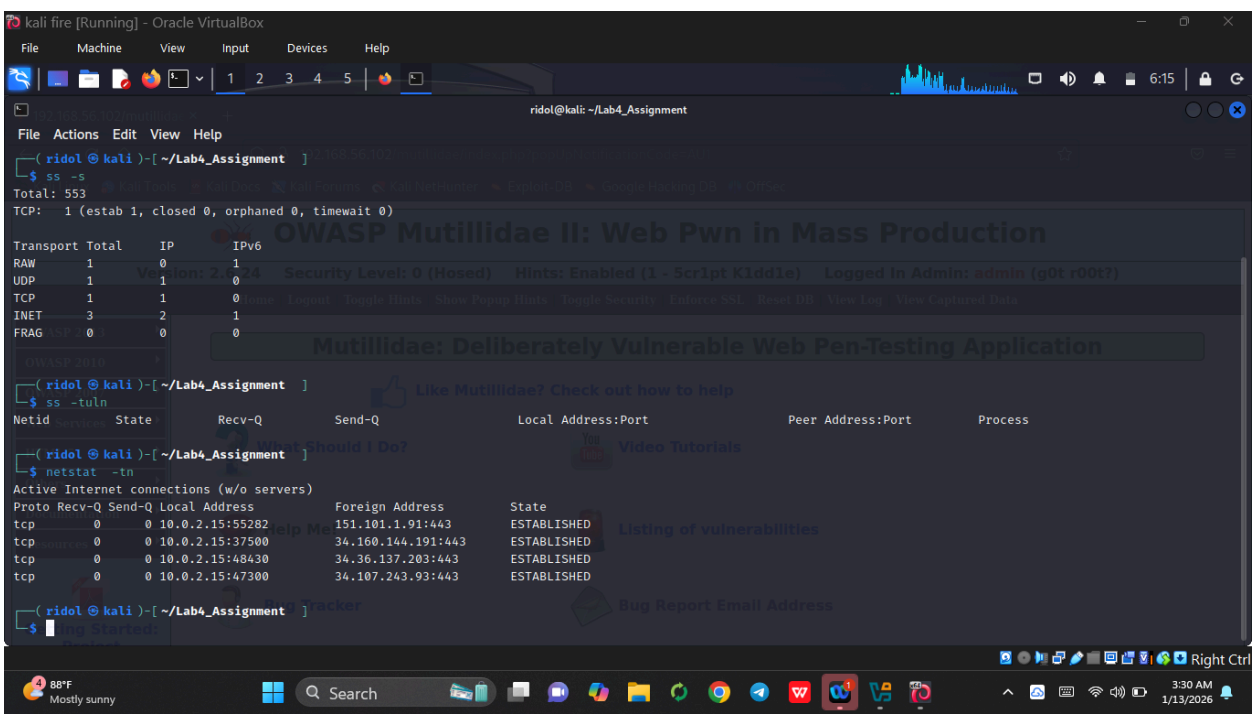

Fig 5:Evidence showing network statistics on the Linux terminal

3.4 Monitoring Network Traffic with Wireshark

- The wireshark was launched from the Linux terminal using the command **wireshark**
- Packet from owaspbwa was captured on the wireshark while some pages were visited on the owaspbwa web server
- **ip.addr == 192.168.56.102** - Filtered traffic all involving 192.168.56.102 IP address
- **tcp.port == 80** - Filtered HTTP traffic
- **tcp.port == 443** - Filtered HTTPS traffic

The traffic was flowing from **10.0.2.15 (source)** to **192.168.56.102 (destination)** using **TCP** protocol. Wireshark captures HTTP data and reveals how it is encapsulated and carried through each TCP/IP layer from the application down to the physical transmission.

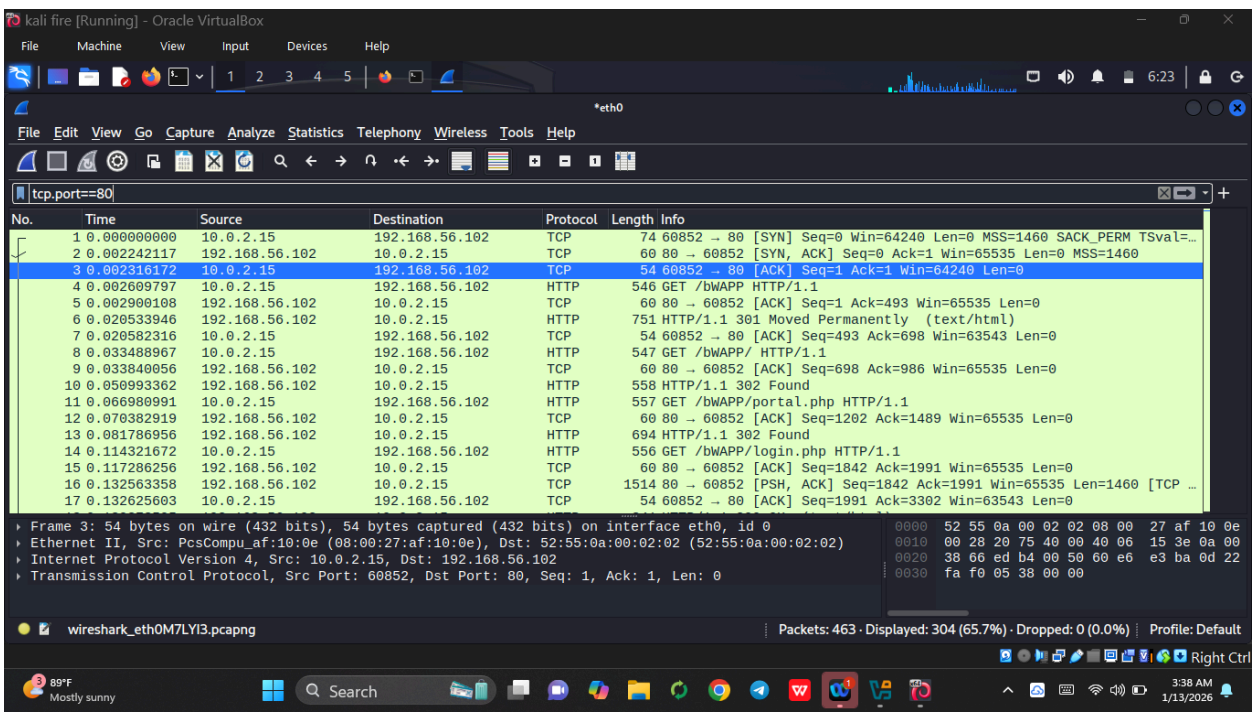

Fig 6: Evidence showing http traffic on wireshark interface

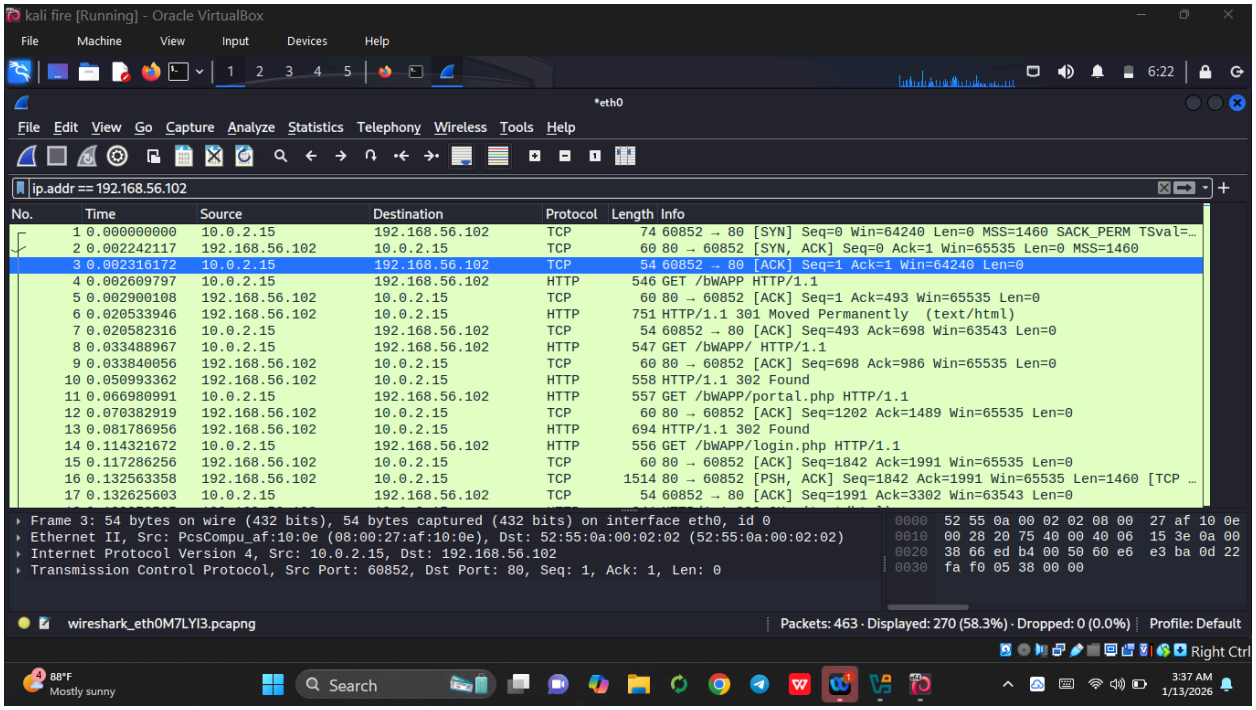

Fig 7: Evidence showing ip address 192.168.56.102 traffic on wireshark interface

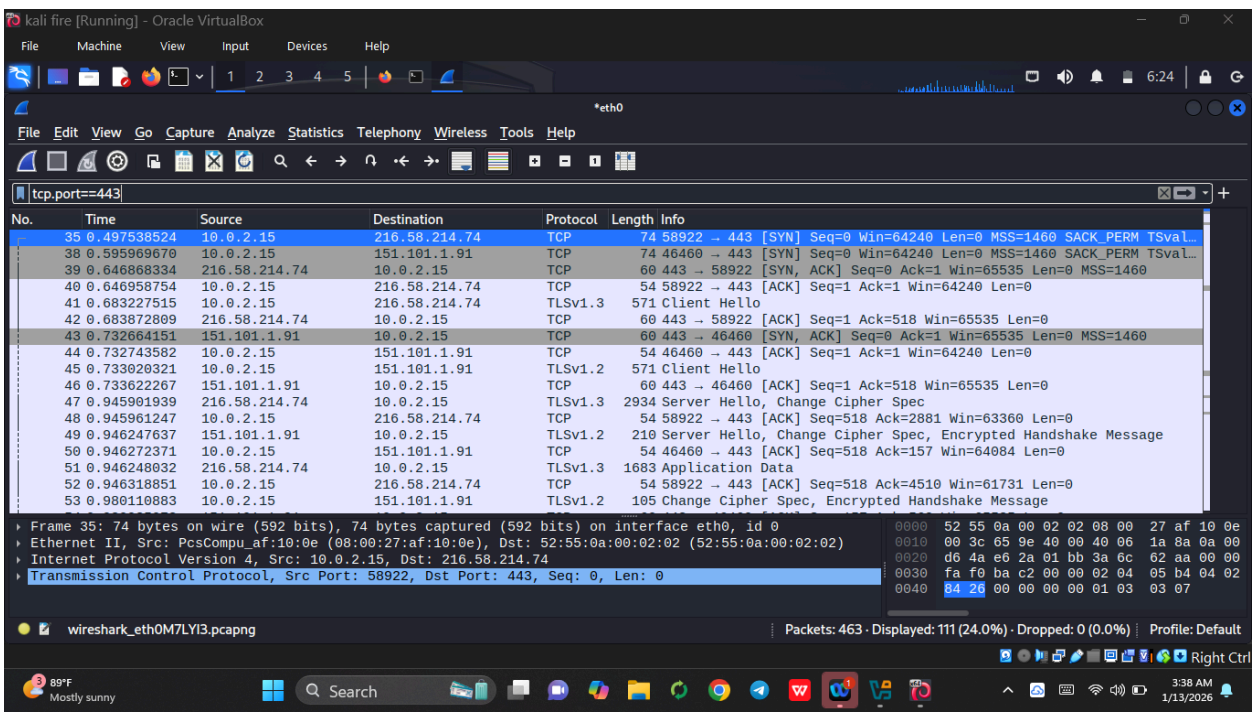

Fig 8: Evidence showing https traffic on wireshark interface

3.5 Packets Transmission Analysis

- The Linux terminal was opened and the following command was run
- **Ping 192.168.56.102** on one terminal and **tcpdump -i eth0 and host 192.168.56.102** on another terminal
- The ping terminal returned an **ICMP reply indicating reachability** while the tcpdump terminal showed a corresponding **echo request and echo reply** to the ping from **source 10.0.2.15**
- No TCP connection was observed at this stage from the ping indicating that this is strictly a **network layer** operation at this stage and **not at transport layer** yet.

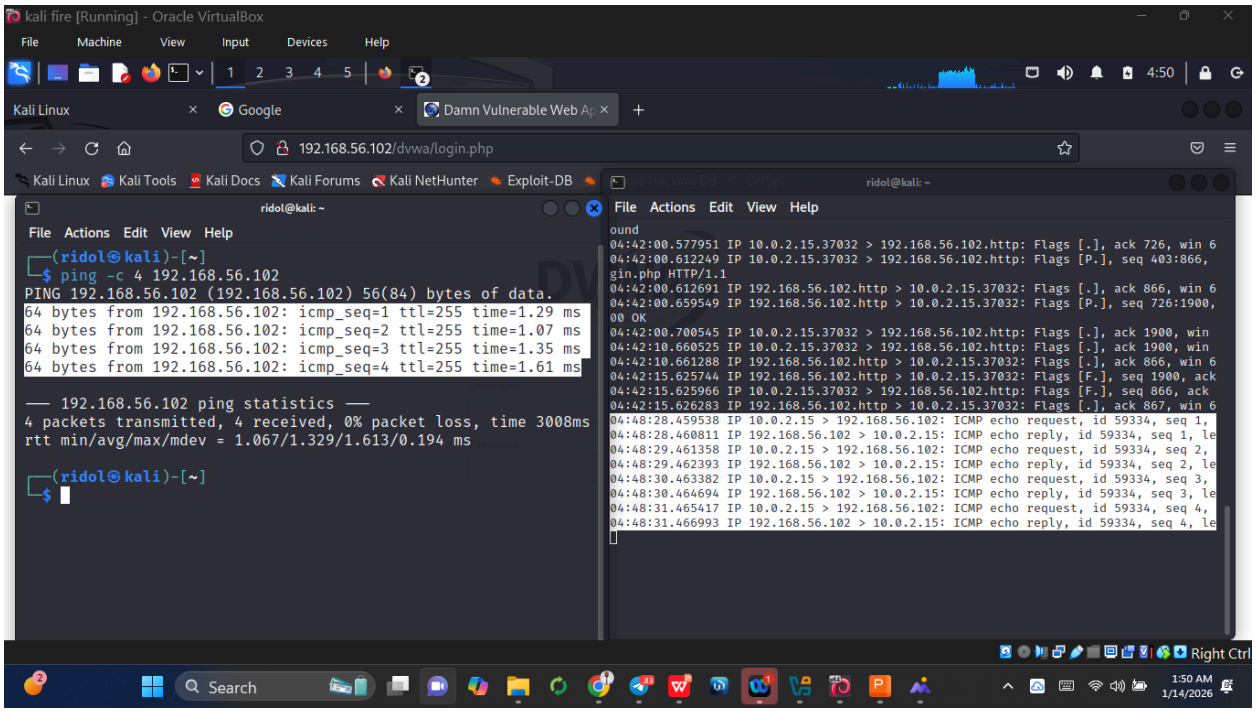

Fig 9: Evidence showing echo request and reply on Linux Terminal

3.6 Troubleshooting Network Interface Issues

- The Linux terminal was opened and the following command was run
- **Sudo ifconfig -i eth0 down** and website was visited
- **Sudo ifconfig -i eth0 up** and website was revisited visited again
- It was observed that the website was **unreachable** in the first visit attempt as a result of interface **eth0 shutdown**

- The website was **reachable** immediately the interface **eth0 was activated** again

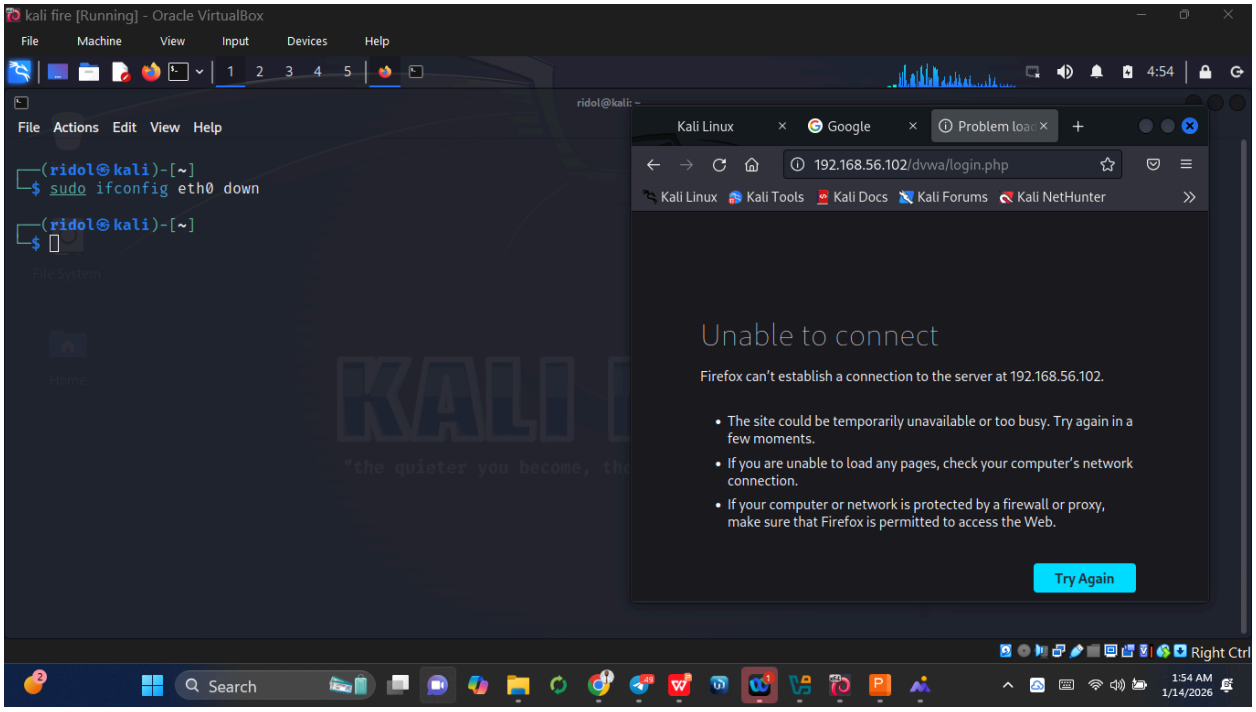

Fig 10: Evidence showing connection downtime due to interface shutdown

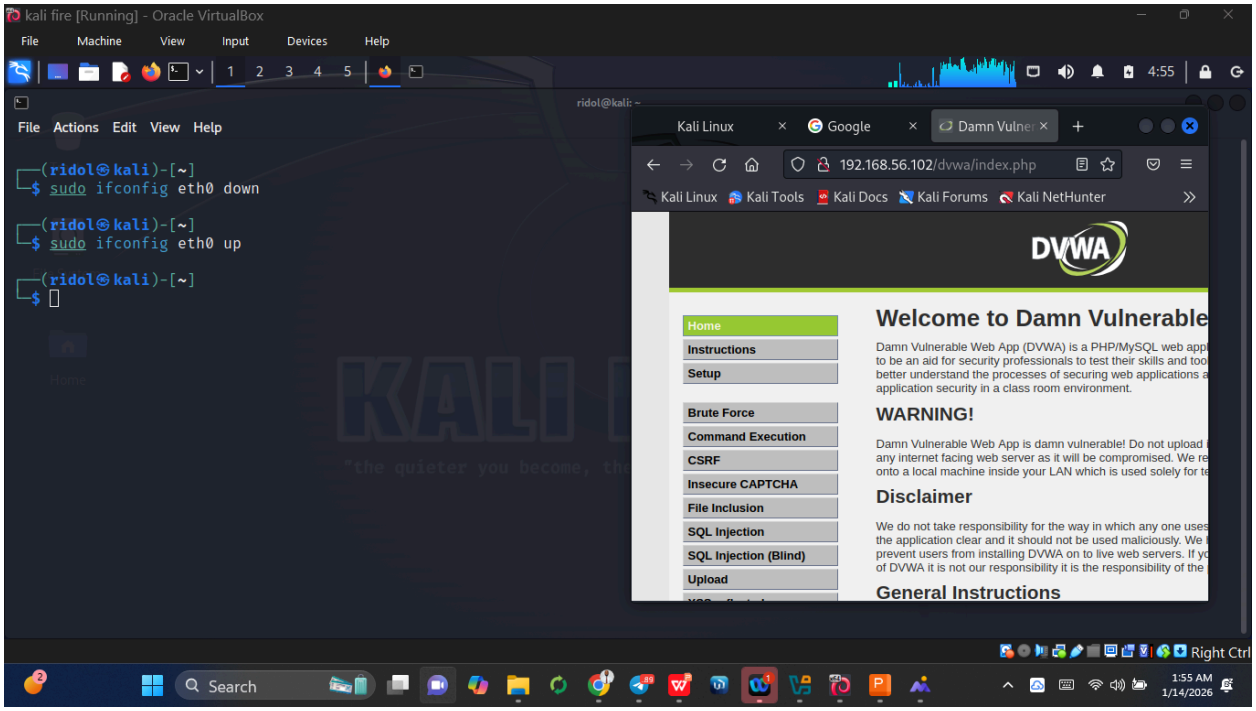

Fig 11: Evidence showing successful connection upon interference reactivation

3.7 Bandwidth Monitoring

- The Linux terminal was opened and the following command was run
- **Iftop -i eth0**
- **nload -t 200 -i 1024 -o 128 -U M**
- This opened up a live network traffic monitoring interface on the linux terminal showing bandwidth usage and direction of traffic flow.

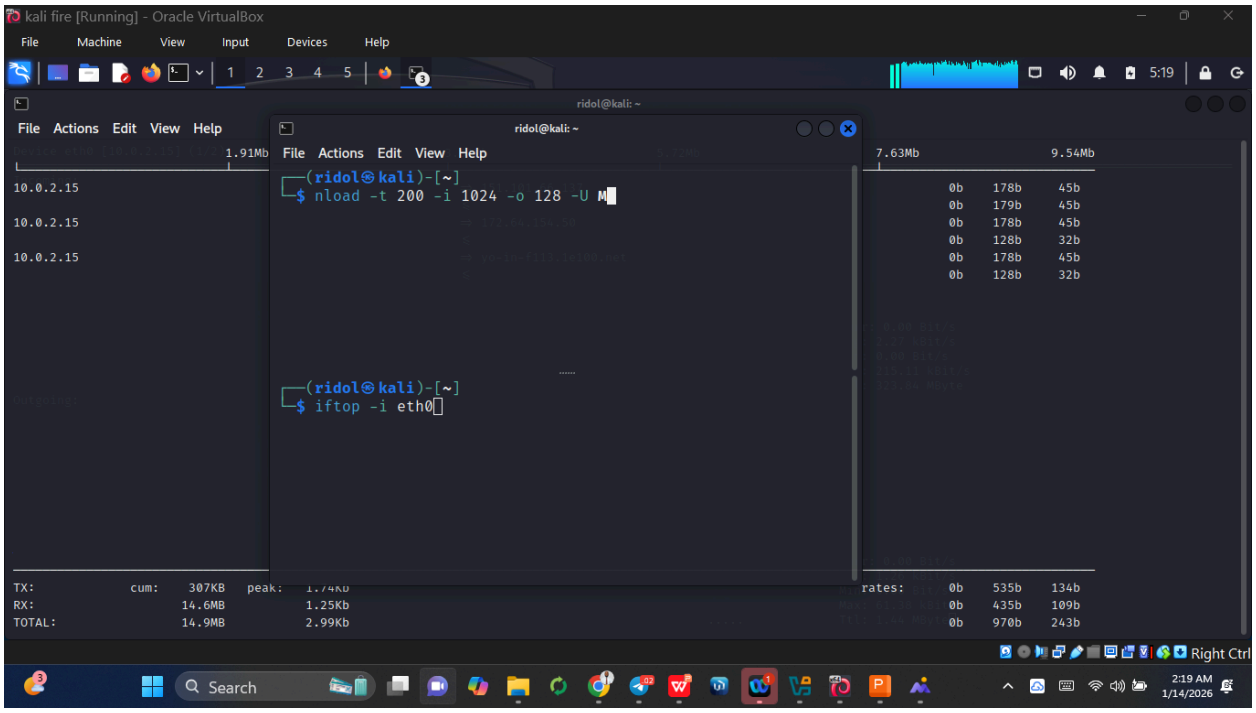

Fig 12: Evidence showing bandwidth usage and traffic flow on Linux terminal

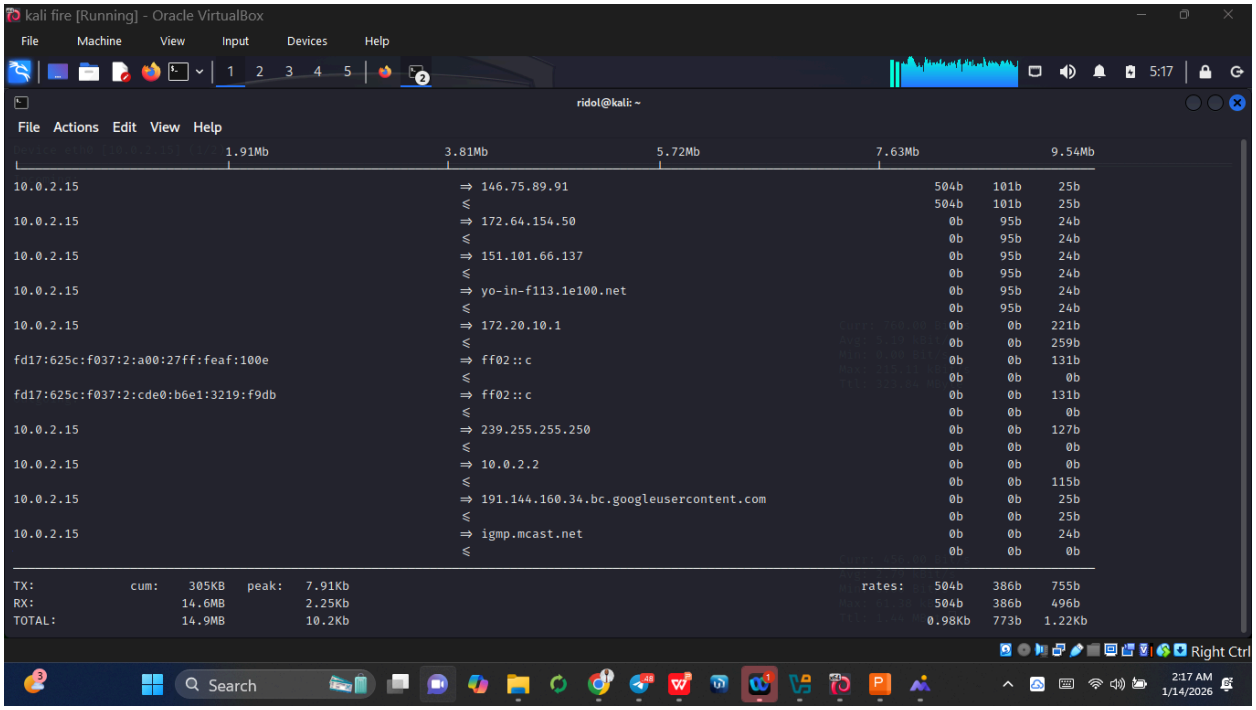

Fig 13: Evidence showing bandwidth usage and traffic flow on Linux terminal

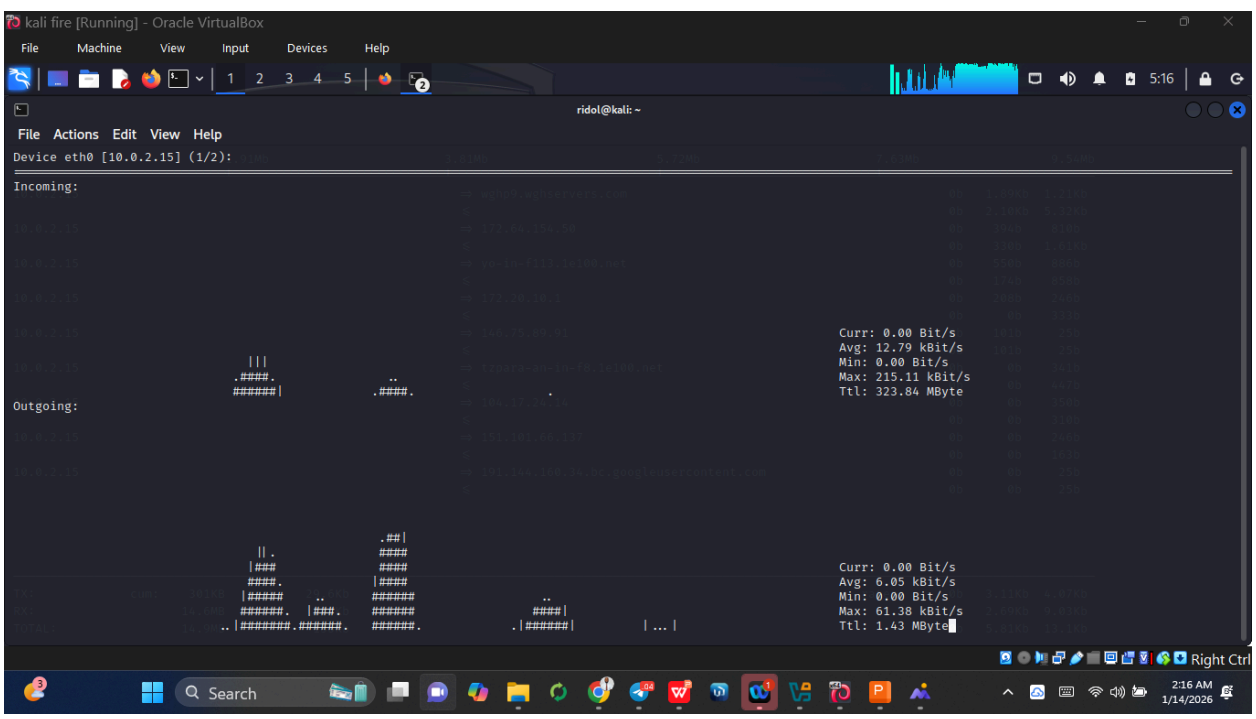

Fig 14: Evidence showing bandwidth usage and traffic flow on Linux terminal

3.8 Advanced Traffic Captured

- The Linux terminal was opened and the following command was run
- **Sudo tcpdump port 80 and host 192.168.56.102**
- This filtered out specific **tcp** traffic to host **192.168.56.102** on **port 80** indicating **http** web traffic

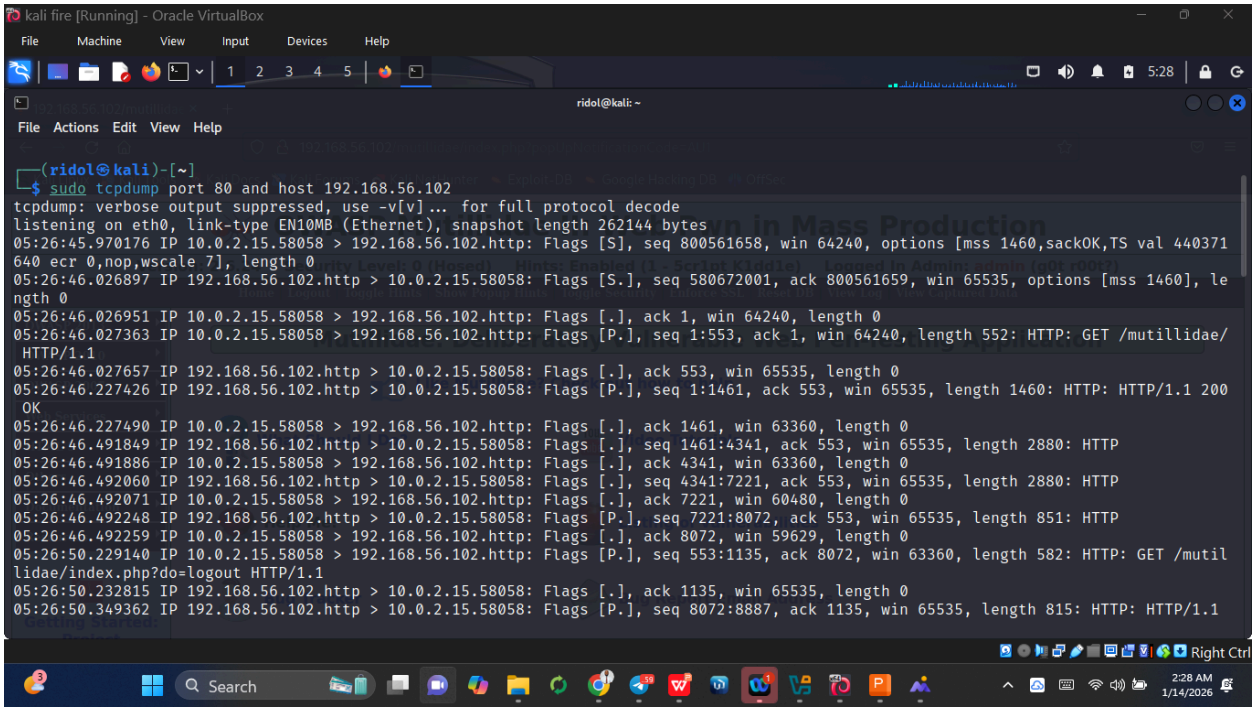

Fig 15: Evidence showing captured traffic flow on Linux terminal

4.0 Results and Analysis

4.1 Network Statistics Analysis

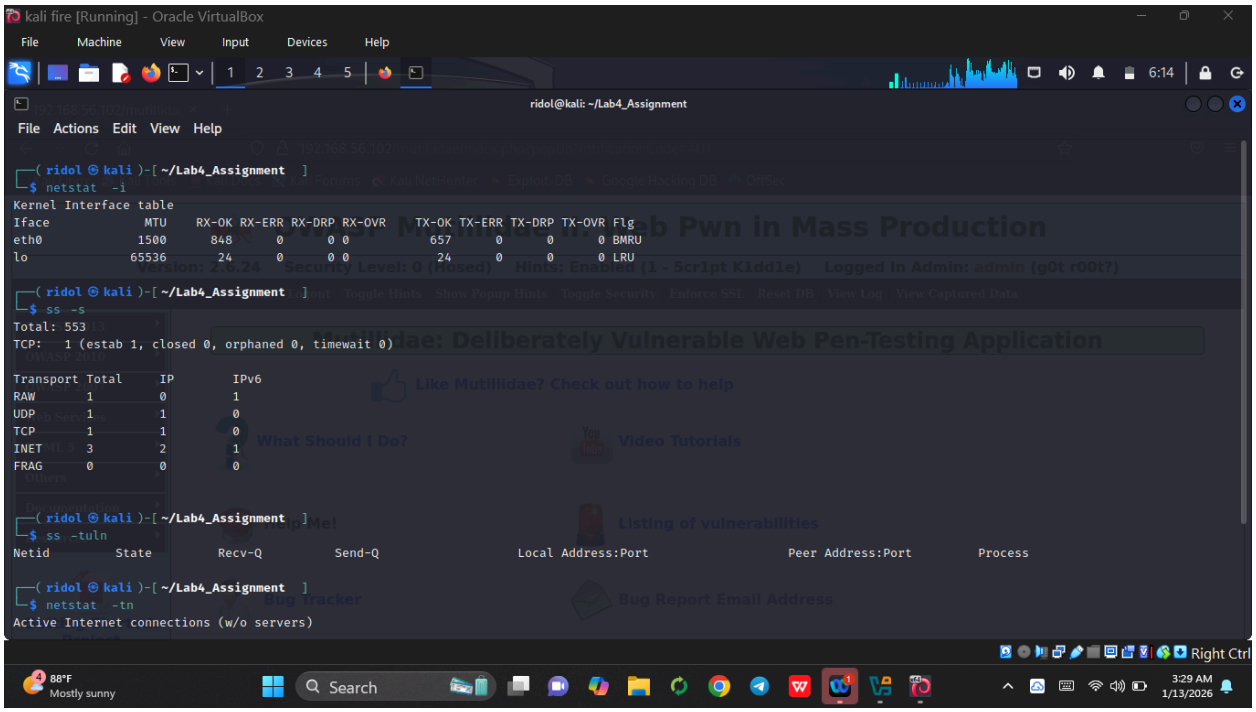

Fig 16:Evidence showing network statistics on the Linux terminal

netstat -i (Kernel Interface Table) displayed the linux network interfaces and traffic statistics.

- **eth0** is the main network interface.
- **RX-OK** and **TX-OK** show received and transmitted packets.
- **RX-ERR, TX-ERR, RX-DRP** indicate errors or dropped packets (all zero here, so the link is healthy).
- **lo** is the loopback interface (used for internal communication within the system).

ss -s (Socket Summary) shows a summary of all sockets. It tells how many **TCP, UDP, RAW**, and other protocol connections exist.

TCP: 1 (estab 1, closed 0, timewait 0) means there is one active TCP connection and no closed or waiting ones.

ss -tuln command lists listening **TCP** and **UDP** ports, but from this lab output, nothing was listening at the moment (no services bound to ports).

netstat -tn shows active **TCP** connections without resolving names. Active Internet connections means there are client connections but no local server processes currently listening.

5.0 Conclusion and Recommendation

The network interfaces serve as the primary gateway for any connection of a host either on the same network or different network; a machine network interface with shutdown status cannot connect or communicate with another machine.

The loopback address serves as a self trust validation in troubleshooting processes to test TCP/IP stack of a machine especially when applying bottom-up approach in a network troubleshooting process.

The transport layer of both OSI model and TCP/IP suite validate and reveal connections between hosts in a sequential manner (SYN, SYN/ACK , ACK). This is very resourceful in tracking legitimate network connection or abuse between host and servers.

It's highly recommended for network administrators and security engineers to secure HTTP traffic across the TCP/IP model, use HTTPS with TLS for encryption, enforce firewall and ACL rules to control IP and port access, segment the network with VLANs to limit sniffing, deploy IDS/IPS to detect abnormal TCP and HTTP activity, and disable unnecessary services and open ports to reduce the attack surface.